OL Organic Letters

pubs.acs.org/OrgLett Letter

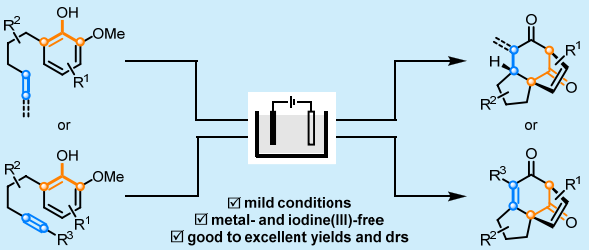
Electrochemical ODI-[5+2] Cascade for the Syntheses of Diversely Functionalized Bicyclo[3.2.1]octane Frameworks

Zhaobo Liu, Jialei Hu,* and Hanfeng Ding*

Cite This: *Org. Lett.* 2021, 23, 6745−6749 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: A metal- and hypervalent iodine reagent-free electrochemical oxidative dearomatization-induced [5+2] cycloaddition/pinacol rearrangement cascade reaction was described. The electrosynthetic method showed strong tolerance for vinylphenols, ethynylphenols, and allenylphenols, which thus enabled the rapid assembly of diversely functionalized bicyclo[3.2.1]octanes in 41−95% yields and up to >20:1 dr. This protocol could be scaled up to gram amounts and should find wide application in complex natural product synthesis.



Downloaded via FUDAN UNIV on January 17, 2022 at 12:10:31 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

Bicyclo[3.2.1]octane is a structurally ubiquitous motif found in various families of biologically active natural products, as represented by terpenoids and alkaloids (Scheme 1a). Due to its indispensable branching-point position leading to the other fused-ring systems, including bicyclo[2.2.2]-, bicyclo[3.3.0]-, and tricyclo[3.2.1.0]octanes, in the biogenetic pathway of tetracyclic diterpenoids and its structure-related prominent bioactivities,[1] the bicyclo[3.2.1]octane scaffold has made itself the subject of intense research and evoked continuing synthetic innovation. To date, numerous elegant methodologies have been disclosed with respect to its construction,[2] which can be chiefly categorized into single-bond cyclizations,[3] multiple-bond cyclizations,[4] and ring-reorganization reactions.[5] Recently, we have described a novel and efficient oxidative dearomatization-induced (ODI) [5+2] cycloaddition[6]/pinacol rearrangement cascade reaction with the aid of a hypervalent iodine(III) reagent, which offers a robust synthetic protocol for the assembly of bicyclo[3.2.1]-octane frameworks from readily available vinylphenols (Scheme 1c).[7] Subsequently, substantial progress in tetracyclic diterpenoid syntheses via ODI-[5+2] cascade reaction has been achieved through our persistent efforts in the past few years.[7a,8] Nevertheless, during the adventures, we found that the non-negligible drawbacks associated with the strongly acidic byproducts derived from the hypervalent iodine reagents greatly limited functional group tolerance, especially for the acid sensitive benzyl alcohol-containing substrates due to the tendency to generate an unstable quinonmethide intermediate.[7a,8c] Therefore, further improvements remain urgently needed in terms of the compatibility between diverse functional groups and the established conditions for the ODI-[5+2] cascade reactions.

Organic electrosynthesis has proven to be a practical technique for the development of novel chemical transformations driven by redox processes with electric current.[9] An obvious advantage of employing the electrochemical methods is that such reactions can be triggered by precise manipulation of the electric current or potential, thus permitting chemists to obtain products that are chemical reagent-based synthetically inaccessible or low-efficient.[10] In 1991, the Yamamura group described an interesting anodic oxidation of vinylphenol (Scheme 1b).[11a] Upon treatment of **1** with a constant current to initiate a two-electron oxidative dearomatization, phenoxy cation **2** was generated in situ, which then reacted with the tethered olefin in a manner of intramolecular [5+2] cycloaddition. The reaction finally delivered tricycle **3** in 59% yield. On the basis of this key transformation, total syntheses of several cedrene-type natural products were accomplished.[11b] However, despite the appealing application potential, the substrate scope of this protocol had not been fully demonstrated. Inspired by the seminal work described above, we herein report an electrochemical ODI-[5+2] cycloaddition/pinacol rearrangement cascade reaction of various types of phenol substrates, including vinylphenols, ethynylphenols, and allenylphenols (Scheme 1c), which exhibited evident advantages over conventional hypervalent iodine oxidation.

We first identified the optimal reaction conditions for the electrochemical ODI-[5+2] cascade by employing vinylphenol **4a** as a model substrate (Table 1; for a detailed optimization, see Table S1), which involve the constant-current (5 mA) electrolysis using a graphite anode and a graphite cathode in an undivided cell containing an electrolyte solution of *n*-Bu_4NBF_4

Received: July 11, 2021
Published: August 17, 2021



© 2021 American Chemical Society

https://doi.org/10.1021/acs.orglett.1c02321
Org. Lett. 2021, 23, 6745−6749

Scheme 1. Construction of the Bicyclo[3.2.1]octane Frameworks

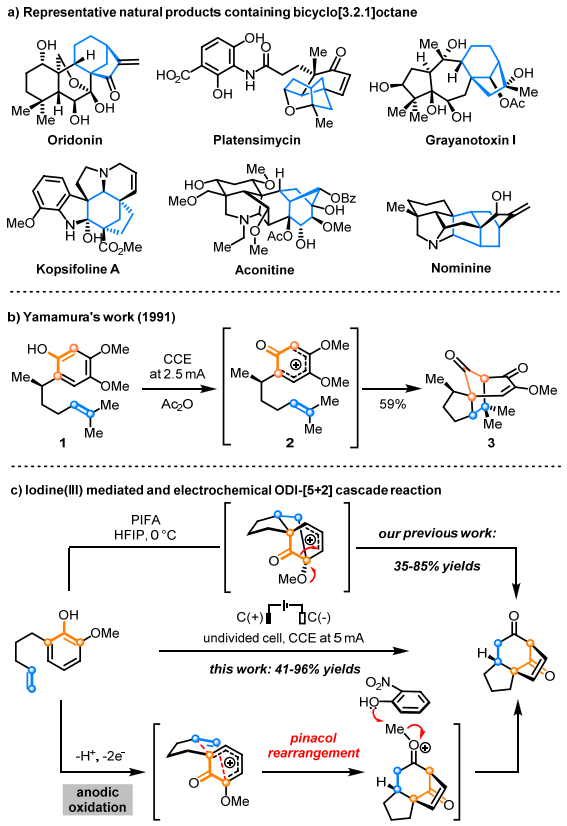
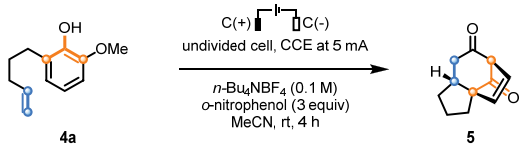


Table 1. Optimization of Reaction Conditions



entry	deviation from the standard conditions	yield (%)[b]
1	none[a]	90 (78)[c]
2	DCE as the solvent	54
3	HFIP as the solvent	25
4	MeOH as the solvent	15
5	n-Bu_4PF_6 as the electrolyte	85
6	2 mA	48
7	10 mA	72
8	HCl (2 M, aqueous) as the additive	25
9	TFA as the additive	84
10	2 equiv of o-nitrophenol	70
11	**4a** (15 mmol)	89[d]

[a]Reaction conditions: undivided cell, graphite anode, graphite cathode, constant current of 5 mA, **4a** (0.20 mmol), o-nitrophenol (0.60 mmol), n-Bu_4NBF_4 (0.1 M), argon, MeCN (10 mL), rt, 4 h. Abbreviations: DCE, 1,2-dichloroethane; HFIP, 1,1,1,3,3,3-hexafluoro-2-propanol; PIFA, [bis(trifluoroacetoxy)iodo]benzene; TFA, trifluoroacetic acid. [b]Isolated yields. [c]Yields in parentheses refer to hypervalent iodine oxidation: PIFA (0.22 mmol), HFIP (10 mL), 0 °C, 20 min. [d]For 50 h.

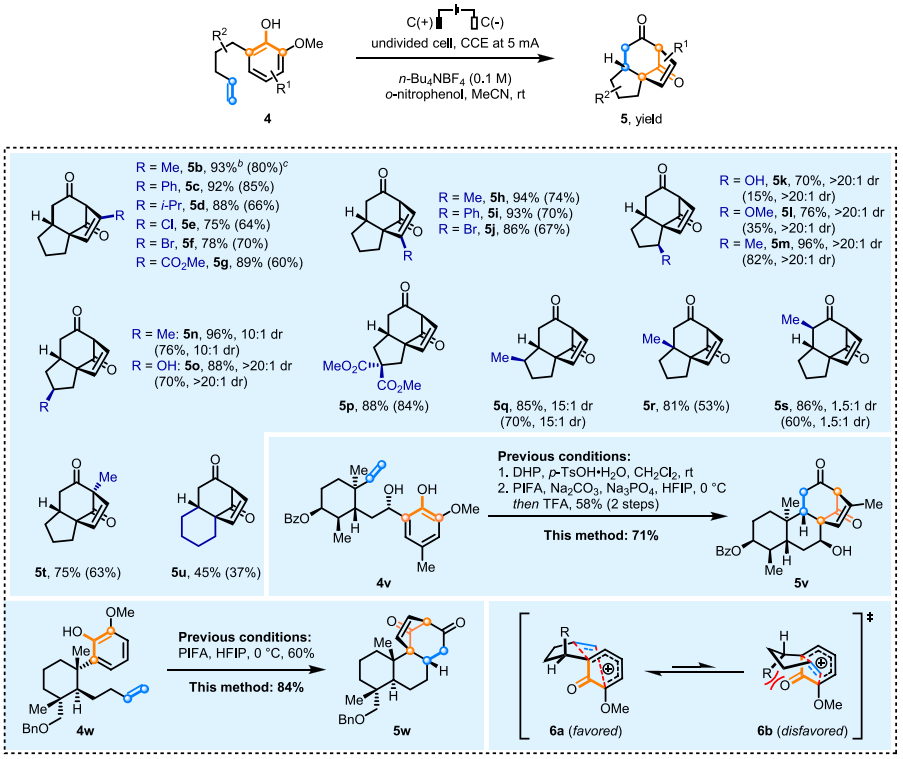
(0.1 M) in MeCN, and o-nitrophenol (3 equiv) as the additive. Under these conditions, the desired bicyclo[3.2.1]octane **5a** was isolated in 90% yield (entry 1, Table 1). Subsequently, switching the solvent to DCE, HFIP, or MeOH resulted in a significantly decreased yield (entries 2−4). Other electrolytes such as n-Bu_4NPF_6 could also be used without affecting the yield or current efficiency (entry 5). A relatively suitable current density was potentially important, because a lower or higher current (2 or 10 mA, respectively) showed a negative influence on the cascade reaction (entries 6 and 7). Because of the generation of benzoquinone byproducts, a reduced efficiency was observed when the reaction mixture was exposed to moisture (entry 8). As a comparison, the replacement of an acid additive under anhydrous conditions has trifling impacts on the yield (entry 9). Attempts to reduce the amount of additive led to a diminished yield and conversion (entry 10). Furthermore, the reaction could be achieved on a gram scale without a decrease in yield, albeit requiring a prolonged reaction time (entry 11). On the other hand, vinylphenols bearing an ethoxy group or a benzyloxy group instead of the methoxy group were also compatible and provided an equal efficiency, whereas a rapid decomposition of the corresponding catechol-type substrate was observed.

Having secured the optimal conditions, we next evaluated the scope of the anodic ODI-[5+2] cascade reaction using an array of vinylphenols with different electronic and steric variations (Scheme 2). The electrochemical protocol proved to be suitable for a wide range of substituents, including alkyl, phenyl, halogens, and esters at various positions of the phenol ring, leading to the corresponding products in good to excellent yields (63−94%, **5b−j** and **5t**). These results indicated that the electronic effects and steric hindrance of the phenol motif had little influence on this reaction. The side chain of the vinylphenol showed a strong tolerance to substituents of broad types, such as methyl (**4m**, **4n**, and **4q**), hydroxyl (**4k** and **4o**), methoxy (**4l**), and *gem*-diester (**4p**), again resulting in high efficiency and excellent diastereoselectivities (84−96%, >10:1 dr). These outcomes presumably arose from minimization of steric interactions between the substituents on the side chain and the carbonyl group in an *exo*-transition state (**6a** vs **6b**). It is worth emphasizing that one major advantage of our method lies in the efficient transformation of the benzylic alcohol-containing substrates in an almost neutral reaction environment, which were previously found to afford the anticipated products in the presence of hypervalent iodine reagents in significantly decreased yields owing to the tendency to generate the unstable quinone methides (**4k** and **4l**). Notably, 1,1- and 1,2-disubstituted terminal alkenes were also viable substrates, providing the desired products with satisfactory yields (81% and 86% for **5r** and **5s**, respectively). Compared with **5a**, cyclohexyl-fused **5u** was delivered with reduced efficiency (45%). The outcome described above was probably attributed to the lack of the Thorpe−Ingold effect[12] in the cycloaddition step. To our delight, **5v**, an advanced intermediate en route to rhodomollanol A, was also achieved from **4v** in 71% yield under the standard conditions. It should be mentioned that the latter was extremely acid-labile. Therefore, a THP preprotection of its benzyl alcohol and addition of an extraneous base to the reaction mixture were required in a previous hypervalent iodine oxidation.[8c] Moreover, **4w** could be transformed into **5w**, a key intermediate of stemarin, in a much improved yield (84%), as well.[8b] These examples clearly implied the future synthetic application prospects of this established protocol.

The electrochemical method was next found to be capable of generating bicyclo[3.2.1]octane products that embedded with

https://doi.org/10.1021/acs.orglett.1c02321
Org. Lett. 2021, 23, 6745−6749

Scheme 2. Scope of Vinylphenols[a]

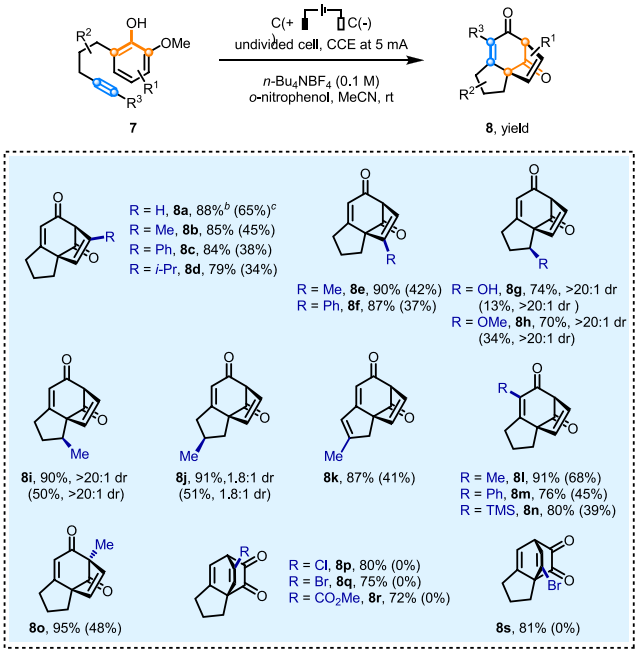


[a]Reaction conditions: undivided cell, graphite anode, graphite cathode, constant current of 5 mA, **4** (0.20 mmol), *o*-nitrophenol (0.60 mmol), *n*-Bu_4NBF_4 (0.1 M), argon, MeCN (10 mL), rt, 4 h. Abbreviation: DHP, dihydropyran. [b]Isolated yields. [c]Yields in parentheses refer to hypervalent iodine oxidation: PIFA (0.22 mmol), HFIP (10 mL), 0 °C, 20 min.

an α,β-unsaturated ketone motif by using ethynylphenols as substrates (Scheme 3). In its present form, the scope of the reaction encompassed ethynylphenols with variable steric substitutions on the phenol ring (**7a**–**f** and **7o**), different functionalities on the side chain (**7g**–**j**), and skipped enynes (**7k**) and even internal alkynes (**7l**–**n**). Accordingly, the expected products were obtained in good to excellent yields (70–95%, **8a**–**o**). Among these experiments, an unexpected double-bond migration was detected with respect to **7k**, which was transformed into a thermodynamically more stable dienone **8k**. Furthermore, in contrast to vinylphenols, we were surprised to find that the substrates bearing electron-withdrawing groups (Cl, Br, CO_2Me, etc.) on the phenol ring (**7p**–**s**) exclusively afforded the Diels–Alder adducts (72–80%, **8p**–**s**), which was probably due to the demethylation to generate an unmasked *o*-benzoquinone prior to [5+2] cycloaddition.[8d]

Encouraged by the aforementioned promising results, we further extended the scope of the reaction to allenylphenols, which appear to be more vulnerable under hypervalent iodine oxidation (Scheme 4). Fortunately, the mild electrochemical conditions successfully overcame the highly acid sensitive nature of these compounds. Greatly improved yields for the formation of **10a**–**p** were realized when compared with those of conventional chemical oxidation. By employing the well-designed electrolytic method, allenylphenols containing both electron neutral and deficient phenol rings could be efficiently and regioselectively converted into the corresponding bicyclo[3.2.1]octanes bearing an exocyclic double bond adjacent to the carbonyl moiety (71–92%, **10a**–**k**). Gratifyingly, substrates bearing distinct substitutions on the side chain

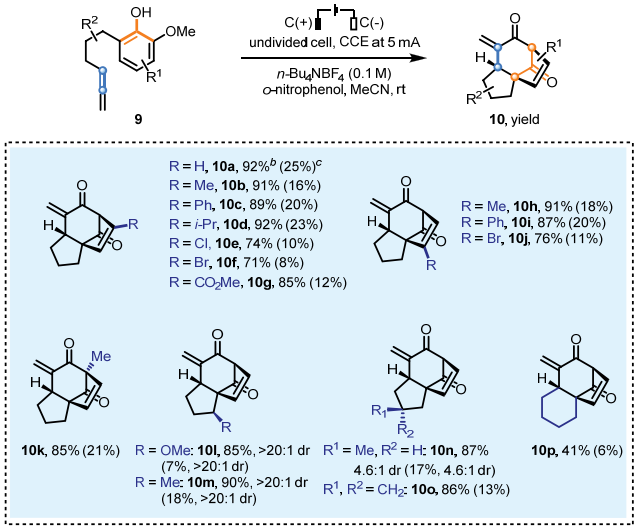
Scheme 3. Scope of Ethynylphenols[a]



[a]Reaction conditions: undivided cell, graphite anode, graphite cathode, constant current of 5 mA, **7** (0.20 mmol), *o*-nitrophenol (0.60 mmol), *n*-Bu_4NBF_4 (0.1 M), argon, MeCN (10 mL), rt, 4 h. [b]Isolated yields. [c]Yields in parentheses refer to hypervalent iodine oxidation: PIFA (0.22 mmol), HFIP (10 mL), 0 °C, 40 min.

https://doi.org/10.1021/acs.orglett.1c02321
Org. Lett. 2021, 23, 6745–6749

Scheme 4. Scope of Allenylphenols[a]



[a]Reaction conditions: undivided cell, graphite anode, graphite cathode, constant current of 5 mA, **9** (0.20 mmol), *o*-nitrophenol (0.60 mmol), n-Bu_4NBF_4 (0.1 M), argon, MeCN (10 mL), rt, 4 h. [b]Isolated yields. [c]Yields in parentheses refer to hypervalent iodine oxidation: PIFA (0.22 mmol), HFIP (10 mL), 0 °C, 20 min.

were also subjected to anodic oxidation, forging the anticipated products in good to excellent yields and satisfactory dr values (41−90%, >4.6:1 dr, **10l−p**).

In conclusion, we have developed a mild electrochemical ODI-[5+2] cycloaddition/pinacol rearrangement cascade reaction. Compared with hypervalent iodine(III)-mediated oxidative conditions, this new protocol significantly improves the reaction efficiency and substrate compatibility, hence providing rapid access to diversely functionalized bicyclo[3.2.1]octanes with good to excellent yields and diastereoselectivities. Its synthetic value has been preliminarily demonstrated by gram-scale preparations of the core frameworks and successful application in the construction of several key intermediates en route to complex natural products. Further studies of application of this methodology are currently underway in our laboratory.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c02321.

Experimental procedures, compound characterization, and computational details (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Hanfeng Ding − *Center of Chemistry for Frontier Technologies, Department of Chemistry, Zhejiang University, Hangzhou 310058, China;* orcid.org/0000-0002-1781-4604; Email: hfding@zju.edu.cn

Jialei Hu − *Center of Chemistry for Frontier Technologies, Department of Chemistry, Zhejiang University, Hangzhou 310058, China;* orcid.org/0000-0002-7506-1612; Email: hujialei@zju.edu.cn

Author

Zhaobo Liu − *Department of Chemistry, Zhejiang University, Hangzhou 310058, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.1c02321

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

Financial support was provided by the National Natural Science Foundation of China (21871230 and 21622205) and the National Ten Thousand Talent Program (W03070091).

■ REFERENCES

(1) (a) Sun, H.-D.; Huang, S.-X.; Han, Q.-B. Diterpenoids from *Isodon* Species and Their Biological Activities. *Nat. Prod. Rep.* **2006**, *23*, 673. (b) Liu, M.; Wang, W.-G.; Sun, H.-D.; Pu, J.-X. Diterpenoids from *Isodon* Species: An Update. *Nat. Prod. Rep.* **2017**, *34*, 1090. (c) Li, C.-H.; Zhang, J.-Y.; Zhang, X.-Y.; Li, S.-H.; Gao, J.-M. An Overview of Grayanane Diterpenoids and Their Biological Activities from the Ericaceae Family in the Last Seven Years. *Eur. J. Med. Chem.* **2019**, *166*, 400.

(2) For reviews of the syntheses of bicyclo[3.2.1]octanes, see: (a) Filippini, M.-H.; Rodriguez, J. Synthesis of Functionalized Bicyclo[3.2.1]octanes and Their Multiple Uses in Organic Chemistry. *Chem. Rev.* **1999**, *99*, 27. (b) Presset, M.; Coquerel, Y.; Rodriguez, J. Syntheses and Applications of Functionalized Bicyclo[3.2.1]octanes: Thirteen Years of Progress. *Chem. Rev.* **2013**, *113*, 525. (c) Zhu, L.; Huang, S.; Yu, J.; Hong, R. Constructive Innovation of Approaching Bicyclo[3.2.1]octane in *ent*-Kauranoids. *Tetrahedron Lett.* **2015**, *56*, 23.

(3) For a recent review, see: (a) Zhao, X.; Cacherat, B.; Hu, Q.; Ma, D. Recent Advances in the Synthesis of *ent*-Kaurane Diterpenoids. *Nat. Prod. Rep.* **2021**, DOI: 10.1039/D1NP00028D. For selected syntheses of bicyclo[3.2.1]octanes, see: (b) Yuan, Z.; Zeng, Y.; Feng, Z.; Guan, Z.; Lin, A.; Yao, H. Constructing Chiral Bicyclo[3.2.1]-octanes via Palladium-Catalyzed Asymmetric Tandem Heck/Carbonylation Desymmetrization of Cyclopentenes. *Nat. Commun.* **2020**, *11*, 2544. (c) Wang, Z.; Martin, S. F. Total Syntheses of (±)-Melicolones A and B. *Org. Lett.* **2020**, *22*, 9071. (d) Wang, G.; Li, J.-C.; Zhou, Y.-G.; Ye, Z.-S. Enantioselective Synthesis of Indole-Fused Bicyclo[3.2.1]octanes via Palladium(II)-Catalyzed Cascade Reaction. *Org. Lett.* **2021**, *23*, 802.

(4) (a) Ellerbrock, P.; Armanino, N.; Ilg, M. K.; Webster, R.; Trauner, D. An Eight-Step Synthesis of Epicolactone Reveals Its Biosynthetic Origin. *Nat. Chem.* **2015**, *7*, 879. (b) Su, X.; Sun, Y.; Yao, J.; Chen, H.; Chen, C. Acid-Promoted Bicyclization of Arylacetylenes to Benzobicyclo[3.2.1]octanes through Cationic Rearrangements. *Chem. Commun.* **2016**, *52*, 4537. (c) Fukazawa, T.; Ando, Y.; Ohmori, K.; Hayashi, T.; Suzuki, K. Enantioselective Access to Bicyclo[3.2.1]octadienone Skeleton: Total Syntheses of (+)-Engelharquinone and Its Epoxide. *Org. Lett.* **2017**, *19*, 1470. (d) Zhu, L.; Ma, W.; Zhang, M.; Lee, M. M.-K.; Wong, W.-Y.; Chan, B. D.; Yang, Q.; Wong, W.-T.; Tai, W. C.-S.; Lee, C.-S. Scalable Synthesis Enabling Multilevel Bio-Evaluations of Natural Products for Discovery of Lead Compounds. *Nat. Commun.* **2018**, *9*, 1283. (e) Richter, M. J. R.; Schneider, M.; Brandstätter, M.; Krautwald, S.; Carreira, E. M. Total Synthesis of (−)-Mitrephorone A. *J. Am. Chem. Soc.* **2018**, *140*, 16704. (f) Zhou, S.; Xia, K.; Leng, X.; Li, A. Asymmetric Total Synthesis of Arcutinidine, Arcutinine, and Arcutine. *J. Am. Chem. Soc.* **2019**, *141*, 13718. (g) Liu, W.; Niu, S.; Zhao, Z.; Yang, S.; Liu, J.; Li, Y.; Fang, X. Divergent Synthesis of Bicyclo[3.2.1]octenes and Cyclohexenes via Catalytic Annulations of Nazarov Reagent and Vinyl 1,2-Diketones. *Org. Lett.* **2020**, *22*, 7572. (h) Anif Pasha, M.;

Peraka, S.; Ramachary, D. B. Catalytic Asymmetric Synthesis of Benzobicyclo[3.2.1]octanes. *Chem. - Eur. J.* **2021**, *27*, 10563.

(5) (a) Lu, P.; Mailyan, A.; Gu, Z.; Guptill, D. M.; Wang, H.; Davies, H. M. L.; Zakarian, A. Enantioselective Synthesis of (−)-Maoecrystal V by Enantiodetermining C−H Functionalization. *J. Am. Chem. Soc.* **2014**, *136*, 17738. (b) Nishiyama, Y.; Yokoshima, S.; Fukuyama, T. Total Synthesis of (−)-Cardiopetaline. *Org. Lett.* **2016**, *18*, 2359. (c) Takatori, K.; Ota, S.; Tendo, K.; Matsunaga, K.; Nagasawa, K.; Watanabe, S.; Kishida, A.; Kogen, H.; Nagaoka, H. Synthesis of Methylenebicyclo[3.2.1]octanol by a Sm(II)-Induced 1,2-Rearrangement Reaction with Ring Expansion of Methylenebicyclo[4.2.0]-octanone. *Org. Lett.* **2017**, *19*, 3763. (d) Kou, K. G. M.; Kulyk, S.; Marth, C. J.; Lee, J. C.; Doering, N. A.; Li, B. X.; Gallego, G. M.; Lebold, T. P.; Sarpong, R. A Unifying Synthesis Approach to the C_{18}-, C_{19}-, and C_{20}-Diterpenoid Alkaloids. *J. Am. Chem. Soc.* **2017**, *139*, 13882. (e) Smith, B. R.; Njardarson, J. T. [2.2.2]- to [3.2.1]-Bicycle Skeletal Rearrangement Approach to the Gibberellin Family of Natural Products. *Org. Lett.* **2018**, *20*, 2993. (f) Hong, B.; Liu, W.; Wang, J.; Wu, J.; Kadonaga, Y.; Cai, P.-J.; Lou, H.-X.; Yu, Z.-X.; Li, H.; Lei, X. Photoinduced Skeletal Rearrangements Reveal Radical-Mediated Synthesis of Terpenoids. *Chem.* **2019**, *5*, 1671. (g) Kamakura, D.; Todoroki, H.; Urabe, D.; Hagiwara, K.; Inoue, M. Total Synthesis of Talatisamine. *Angew. Chem., Int. Ed.* **2020**, *59*, 479. (h) Xu, Z.-J.; Zong, Y.; Qiao, Y.-N.; Zhang, J.-Z.; Liu, X.; Zhu, M.-Z.; Xu, Y.; Zheng, H.; Fang, L.; Wang, X.-n.; Lou, H.-X. Divergent Total Synthesis of Euphoranginol C, Euphoranginone D, *ent*-Trachyloban-3β-ol, *ent*-Trachyloban-3-one, Excoecarin E, and *ent*-16α-Hydroxy-atisane-3-one. *Angew. Chem., Int. Ed.* **2020**, *59*, 19919.

(6) For reviews of [5+2] cycloaddition, see: (a) Pellissier, H. Recent Developments in the [5 + 2] Cycloaddition. *Adv. Synth. Catal.* **2011**, *353*, 189. (b) Ylijoki, K. E. O.; Stryker, J. M. [5 + 2] Cycloaddition Reactions in Organic and Natural Product Synthesis. *Chem. Rev.* **2013**, *113*, 2244. (c) Pellissier, H. Recent Developments in the [5 + 2] Cycloaddition. *Adv. Synth. Catal.* **2018**, *360*, 1551. (d) Liu, X.; Hu, Y.-J.; Fan, J.-H.; Zhao, J.; Li, S.; Li, C.-C. Recent Synthetic Studies towards Natural Products via [5 + 2] Cycloaddition Reactions. *Org. Chem. Front.* **2018**, *5*, 1217. (e) Gao, K.; Zhang, Y.-G.; Wang, Z.; Ding, H. Recent Development on the [5 + 2] Cycloadditions and Their Application in Natural Product Synthesis. *Chem. Commun.* **2019**, *55*, 1859. (f) Min, L.; Liu, X.; Li, C.-C. Total Synthesis of Natural Products with Bridged Bicyclo[m.n.1] Ring Systems via Type II [5 + 2] Cycloaddition. *Acc. Chem. Res.* **2020**, *53*, 703.

(7) (a) He, C.; Hu, J.; Wu, Y.; Ding, H. Total Syntheses of Highly Oxidized *ent*-Kaurenoids Pharicin A, Pharicinin B, 7-*O*-Acetylpseurata C, and Pseurata C: A [5 + 2] Cascade Approach. *J. Am. Chem. Soc.* **2017**, *139*, 6098. (b) He, C.; Bai, Z.; Hu, J.; Wang, B.; Xie, H.; Yu, L.; Ding, H. A Divergent [5 + 2] Cascade Approach to Bicyclo[3.2.1]-octanes: Facile Synthesis of *ent*-Kaurene and Cedrene-Type Skeletons. *Chem. Commun.* **2017**, *53*, 8435.

(8) For a recent review, see: (a) Gao, K.; Hu, J.; Ding, H. Tetracyclic Diterpenoid Synthesis Facilitated by ODI-Cascade Approaches to Bicyclo[3.2.1]octane Skeletons. *Acc. Chem. Res.* **2021**, *54*, 875. For preliminary application of the ODI-[5+2] cascade in natural product synthesis, see: (b) Hu, J.; Jia, Z.; Xu, K.; Ding, H. Total Syntheses of (+)-Stemarin and the Proposed Structures of Stemara-13(14)-en-18-ol and Stemara-13(14)-en-17-acetoxy-18-ol. *Org. Lett.* **2020**, *22*, 1426. (c) Gao, J.; Rao, P.; Xu, K.; Wang, S.; Wu, Y.; He, C.; Ding, H. Total Synthesis of (−)-Rhodomollanol A. *J. Am. Chem. Soc.* **2020**, *142*, 4592. (d) Wang, B.; Liu, Z.; Tong, Z.; Gao, B.; Ding, H. Asymmetric Total Syntheses of 8,9-Seco-*ent*-kaurane Diterpenoids Enabled by An Electrochemical ODI-[5 + 2] Cascade. *Angew. Chem., Int. Ed.* **2021**, *60*, 14892.

(9) For recent reviews, see: (a) Francke, R.; Little, R. D. Redox Catalysis in Organic Electrosynthesis: Basic Principles and Recent Developments. *Chem. Soc. Rev.* **2014**, *43*, 2492. (b) Yan, M.; Kawamata, Y.; Baran, P. S. Synthetic Organic Electrochemical Methods Since 2000: On the Verge of a Renaissance. *Chem. Rev.* **2017**, *117*, 13230. (c) Möller, K. D. Using Physical Organic Chemistry to Shape the Course of Electrochemical Reactions. *Chem. Rev.* **2018**, *118*, 4817. (d) Yuan, Y.; Lei, A. Electrochemical Oxidative Cross-Coupling with Hydrogen Evolution Reactions. *Acc. Chem. Res.* **2019**, *52*, 3309. (e) Siu, J. C.; Fu, N.; Lin, S. Catalyzing Electrosynthesis: A Homogeneous Electrocatalytic Approach to Reaction Discovery. *Acc. Chem. Res.* **2020**, *53*, 547. (f) Wang, F.; Stahl, S. S. Electrochemical Oxidation of Organic Molecules at Lower Overpotential: Accessing Broader Functional Group Compatibility with Electron−Proton Transfer Mediators. *Acc. Chem. Res.* **2020**, *53*, 561.

(10) For recent reviews, see: (a) Horn, E. J.; Rosen, B. R.; Baran, P. S. Synthetic Organic Electrochemistry: An Enabling and Innately Sustainable Method. *ACS Cent. Sci.* **2016**, *2*, 302. (b) Mohle, S.; Zirbes, M.; Rodrigo, E.; Gieshoff, T.; Wiebe, A.; Waldvogel, S. R. Modern Electrochemical Aspects for the Synthesis of Value-Added Organic Products. *Angew. Chem., Int. Ed.* **2018**, *57*, 6018.

(11) (a) Yamamura, S.; Shizuri, Y.; Shigemori, H.; Okuno, Y.; Ohkubo, M. Natural Products Syntheses Using Anodic Oxidation of Phenols as a Key Step. *Tetrahedron* **1991**, *47*, 635. (b) For a comprehensive review and references therein, see: Yamamura, S.; Nishiyama, S. Anodic Oxidation of Phenols Towards the Synthesis of Bioactive Natural Products. *Synlett* **2002**, *200240533*

(12) For recent reviews, see: (a) Jung, M. E.; Piizzi, G. *gem*-Disubstituent Effect: Theoretical Basis and Synthetic Applications. *Chem. Rev.* **2005**, *105*, 1735. (b) Zheng, Y.; Xu, J. Thorpe-Ingold Effect and Its Application in Cyclizations in Organic Chemistry. *Prog. Chem.* **2014**, *26*, 1471.

https://doi.org/10.1021/acs.orglett.1c02321
Org. Lett. 2021, 23, 6745−6749